Oscillation

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By simple oscillation, we mean the periodic oscillation of a quantity with respect to an equilibrium value. Note that simple oscillation is not merely restricted to the position of an object. The electric field (in the case of light), the angle with the equilibrium position (in a simple pendulum), etc., any quantity can exhibit simple oscillation. For this reason, simple oscillation is very common in physics. Actually, there are many different opinions regarding the characteristics of oscillatory motion. Therefore, we will not go into the discussion of whether the motion of a system is simple harmonic motion or not; our main goal will be to determine how the system changes with time. Generally speaking, if a quantity x obeys a specific equation

$$\ddot{x} + \omega_0^2 x = 0$$

then we say the change of the quantity is simple harmonic. For example, if the quantity x indicates the position of an object, then if the force acting on the object is proportional to its position and acts in the opposite direction of the displacement, the motion of that object will be simple harmonic motion. That is, if $\boldsymbol{F} = -a\boldsymbol{x}$, then

$$m\ddot{x} + ax = 0$$

$$\Rightarrow \ddot{x} + \frac{a}{m}x = 0$$

Here if a is positive, it will represent the equation of simple harmonic motion, whose $\omega_0^2 = \frac{a}{m}$. This $\omega_0 = 2\pi f = \frac{2\pi}{T}$ indicates the angular frequency of the object's motion and T is the time period of the object's motion. Before going into detailed discussion, let us learn how to solve some differential equations.

1 Differential Equations

In any differential equation, the highest derivative of the variable present determines the order of the differential equation. The equation of simple harmonic motion we saw earlier is a second-order differential equation because it contains up to the second derivative of position. It should also be mentioned that the equations we will work with will all be ordinary and linear differential equations. This means that in the equations, all derivatives will have a power of one, that is there will be no terms like $\ddot{x}^2, \dot{x}^2, x^2$. These types of differential equations can again be divided into two categories; homogeneous and inhomogeneous. An equation will be homogeneous if every term in it contains the variable (in this case x) or any of its derivatives. If not, we will call that equation inhomogeneous. Any n-th order linear, homogeneous differential equation has n solutions. The linear combination of these n solutions is the general solution of the differential equation.

1.1 $\ddot{x} + \omega_0^2 x = 0$

Since the equation is quadratic (second-order) and homogeneous, it will have two independent solutions. However we do it, once we get two solutions, we can say that our work is done. You can actually solve this equation by separating the variables and integrating, which you learned earlier in the kinematics chapter. However, let that remain as an exercise. We will learn a more general method, which can also be used to solve other equations of this type.

First, let us look at the equation a bit more carefully. The equation can be written as $\ddot{x} = -\omega_0^2 x$. So, we are looking for a function x, which when differentiated twice gives (a constant $\cdot$ that function). Does any such function come to mind? Differentiating $e^{\alpha t}$ with respect to time will give the product of a constant and the original function! So let us assume our desired function is $x = e^{\alpha t}$. Substituting this x into the differential equation gives,

$$\alpha^2 e^{\alpha t} + \omega_0^2 e^{\alpha t} = 0$$
$$\Rightarrow \alpha^2 + \omega_0^2 = 0$$
$$\Rightarrow \alpha = \sqrt{-\omega_0^2} = \pm i\omega_0$$

We got two values for α; this means our solutions are $x = e^{i\omega_0 t}$ and $x = e^{-i\omega_0 t}$.[1] We have found two mutually independent solutions of the quadratic equation, which means we have found all the solutions.[2] However, we want the general solution. The general solution will be the linear combination of the obtained solutions. It will be easier to show what Linear Combination means rather than explaining it. We will say the general solution of the equation is

$$x = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$$

where A and B are two unknown constants, which can be either complex or real. Note that in the solution of any n-th order differential equation, n unknown constants remain, which depend on the initial state of the variable.

We have solved the equation very easily. We will see in the next section how your familiar sine and cosine functions come from this solution. Now let us discuss the general rule once again. If the coefficients of any second-order, linear, ordinary differential equation are constants, its solution is an exponential function. That is why we assume the solution $x = e^{\alpha t}$ and substitute it into the original equation. For this, we get a quadratic equation for α, solving which we get two values (possibly repeated) of α. For these two α's, x gives our two mutually independent solutions of the equation. By multiplying them with two different constants and adding them together, the general solution of the equation can be obtained.

1.2 $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$

As before, let us assume the solution $x = e^{\alpha t}$. Substituting this into the original equation will give-

$$\alpha^2 e^{\alpha t} + 2\gamma\alpha e^{\alpha t} + \omega_0^2 e^{\alpha t} = 0$$

[1] Some of you are surely wondering, we were representing displacement by x. How did its solution become an imaginary number! Actually, we are only learning to solve the differential equation here. Here x can represent anything, so it can be imaginary too. When x represents displacement or any real variable, we will work with the real part of this solution. You will find a detailed discussion on this later.

[2] What if the two roots of our obtained quadratic equation were equal? Then we would only get one independent solution. How to find the other one, we will come back to that in section 5.3.

$$\Rightarrow \alpha^2 + 2\gamma\alpha + \omega_0^2 = 0$$

Solving the obtained quadratic equation gives

$$\alpha = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} = -\gamma \pm i\sqrt{\omega_0^2 - \gamma^2}$$

The reason for taking i common at the end is that usually $\omega_0 > \gamma$. Then the general solution can be written as

$$x = Ae^{(-\gamma + i\sqrt{\omega_0^2 - \gamma^2})t} + Be^{(-\gamma - i\sqrt{\omega_0^2 - \gamma^2})t}$$

$$\Rightarrow x = e^{-\gamma t}(Ae^{i\sqrt{\omega_0^2 - \gamma^2}t} + Be^{-i\sqrt{\omega_0^2 - \gamma^2}t})$$

1.3 $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f(t)$

The difference between this equation and the previous two equations is that it is not homogeneous. In any inhomogeneous n-th order equation, along with n independent solutions, there remains one additional particular solution (in this case, since there is one inhomogeneous term, there is one extra). We have actually already determined the n independent solutions! Substituting the x obtained in the previous section makes the left side of the equation zero. To this, the additional particular solution will be added, resulting in our left side becoming $f(t)$. Solving the equation will make the matter a bit clearer. We will mainly solve the equation for $f(t) = f_0 e^{i\omega_d t}$, where ω_d is a constant. In solving this equation, we have to think about what the equation actually means in reality. We will return to section 6.

2 Formulating the Differential Equation of Simple Harmonic Motion

In this part, we will return to physics again. Here we will learn two ways to determine the differential equation of motion of any system. In the last part, we will return to solving those equations again.

2.1 Force Method

Just by looking at the name, you might have understood that in this method we will mainly determine the equation of motion using Newton's second law. First, we will work with cases where there is no resistive force. In such cases, if the applied force is proportional to the object's position (x) and is in the opposite direction, then simple harmonic motion is observed. The most common example is a block attached to a spring; expressing the object's position from the equilibrium position by x,

$$F = -kx$$

$$\Rightarrow m\ddot{x} + kx = 0$$

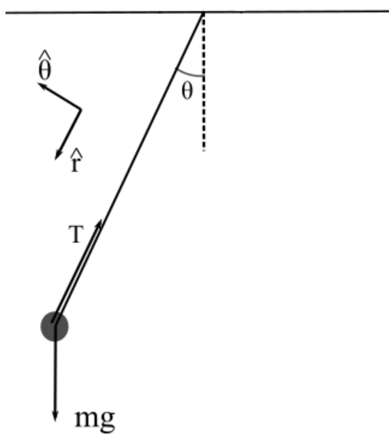
$$\Rightarrow \ddot{x} + \omega_0^2 x = 0$$

where $\omega_0 = \sqrt{\frac{k}{m}}$. The obtained equation is the equation of simple harmonic motion. The reason this example is one of the easiest is because for any x, the restoring force is proportional to x[3]. However, simple harmonic motion is also possible under the influence of many other

[3]That is, of course, within the elastic limit of the spring. The elastic limit of a normal spring is quite large.

forces, but in those cases, simple harmonic motion is only possible for small displacements. If you have little idea about the Taylor series, read the Taylor series section from the appendix right now.

First, we determine the differential equation of motion of a simple pendulum. Suppose a point mass m is attached to a massless string of length L. The other end of the string is connected to a rigid support and the mass is allowed to swing freely. In this case, to apply Newton's law, it is more convenient to use polar coordinates taking the fixed end of the string as the center. The reason is that in this case there is no acceleration along the radius, so we only need to write the equation along its perpendicular direction. If written in Cartesian coordinates, both x-y would change, resulting in the problem becoming two-dimensional. In polar coordinates, only θ is changing, so the problem has become one-dimensional.



Applying Newton's law,

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -mg\sin\theta$$

where we have taken the direction of θ as in the figure above. Remember, if a force tries to increase the value of θ, then F_θ is positive. In this case, since the force is trying to decrease the value of θ, the force has been considered negative. Since the mass's position is $r = L$ and it is not changing, $\dot{r} = 0$. Then-

$$L\ddot{\theta} = -g\sin\theta$$

$$\Rightarrow \ddot{\theta} + \frac{g}{L}\sin\theta = 0$$

which is not the equation of simple harmonic motion. However, we can write the $\sin\theta$ function as $\sin\theta \approx \theta$ for small θ. Basically, because θ is small, we have assumed the terms of its higher powers to be zero. You can plug in values to see for yourself, within $-0.1 < \theta < 0.1$ radians, $\sin\theta \approx \theta$ is fairly true. Even going up to 0.2 radians, $\sin\theta$ and θ remain quite close. So for small θ, it can be written,

$$\ddot{\theta} + \frac{g}{L}\theta = 0$$

which is the equation of simple harmonic motion. So let's review together the rules for determining the differential equation of simple harmonic motion of an object through force:

1. First, the equilibrium state of the object must be determined. Suppose the equilibrium point is q_0 (q can be any coordinate including x, y, θ).

2. Now we calculate the force acting on the object due to a displacement q from equilibrium. That is, if the object's position is $q_0 + q$, we will apply Newton's law on the object and determine its differential equation of motion. Note that there will be no coordinate other than q in the equation. If there are other coordinates, they must usually be replaced by some conservation law to bring everything into the variable q.

3. If the obtained forces are proportional to the displacement q from equilibrium, then our work is done. If not, we will expand the acting force function with respect to q_0 and keep only up to the term containing q. The obtained equation will be the differential equation of simple harmonic motion.

Note here, if the force function is $F(q_0+q)$, its expansion up to the first power term will be

$$F(q_0+q) \approx F(q_0) + F'(q_0)q$$

However, if q_0 is the equilibrium point, $F(q_0) = 0$ will be true.

Example: If the earth is given a slight nudge along its radius, will its motion be simple harmonic? Given that the masses of the sun and the earth are M and m respectively, the earth's orbit is circular, whose radius is R_0.

Solution: We know that at $r = R_0$ the motion along the earth's radius is in equilibrium; since no extra force is being applied, the equilibrium point will remain this. Suppose after giving the nudge the distance of the earth from the sun is $r = R_0 + x$. Applying Newton's law along the radius,

$$m(\ddot{r} - \dot{\theta}^2 r) = -\frac{GMm}{r^2}$$

Since the nudge was given along the radius, the applied force was also along the radius. Therefore, the angular momentum of the earth will be conserved. Then

$$L = mv_\theta r = mv_0 R_0$$

We know, the velocity of an object moving in a circular path of radius R_0 is $v = \sqrt{\frac{GM}{R_0}}$. The angular component of speed $v_\theta = \dot{\theta} r$. Altogether,

$$m\dot{\theta}r^2 = m\sqrt{\frac{GM}{R_0}}R_0$$

$$\Rightarrow \dot{\theta}^2 r = \frac{GMR_0}{r^3}$$

Substituting this,

$$\ddot{r} - \frac{GMR_0}{r^3} + \frac{GM}{r^2} = 0$$

Since the obtained equation is not simple harmonic motion, we have to try using some approximations. First let's substitute $r = R_0 + x$-

$$\ddot{r} - \frac{GMR_0}{(R_0+x)^3} + \frac{GM}{(R_0+x)^2} = 0$$

$$\Rightarrow \ddot{r} - \frac{GMR_0}{R_0^3(1+\frac{x}{R_0})^3} + \frac{GM}{R_0^2(1+\frac{x}{R_0})^2} = 0$$

$$\Rightarrow \ddot{r} - \frac{GM}{R_0^2}\left(\frac{1}{(1+\frac{x}{R_0})^3} - \frac{1}{(1+\frac{x}{R_0})^2}\right) = 0$$

If $x \ll R_0$,

$$\frac{1}{(1+\frac{x}{R_0})^3} \approx 1 - \frac{3x}{R_0}$$

and,

$$\frac{1}{(1+\frac{x}{R_0})^2} \approx 1 - \frac{2x}{R_0}$$

Besides, $\ddot{r} = \ddot{x}$. Substituting the values,

$$\ddot{r} - \frac{GM}{R_0^2}\left(1 - \frac{3x}{R_0} - 1 + \frac{2x}{R_0}\right) = 0$$

$$\Rightarrow \ddot{x} + \frac{GM}{R_0^3}x = 0 \tag{1}$$

which is an equation of simple harmonic motion.

2.2 Energy Method

Suppose the total energy of an object depends only on a single generalized coordinate q. In that case, it can be written in the form

$$E = f(\dot{q}) + g(q)$$

Here we call the function of $\dot{q}$ the generalized kinetic energy and the function of q the generalized or effective potential energy. Think about a spring. In this case, the total energy is

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$$

If energy is conserved, the equation of motion can be determined directly from this energy equation. Differentiating both sides of the energy equation with respect to time

$$\frac{dE}{dt} = \frac{1}{2}m\frac{d\dot{x}^2}{d\dot{x}}\frac{d\dot{x}}{dt} + \frac{1}{2}k\frac{dx^2}{dx}\frac{dx}{dt}$$

where we have used the chain rule. If energy is conserved, $\frac{dE}{dt} = 0$. Therefore,

$$0 = \frac{1}{2}m\frac{d\dot{x}^2}{d\dot{x}}\ddot{x} + \frac{1}{2}k\frac{dx^2}{dx}\dot{x}$$

$$\Rightarrow m\dot{x}\ddot{x} + kx\dot{x} = 0$$

$$\Rightarrow \dot{x}(m\ddot{x} + kx) = 0$$

Now, for most systems, $\dot{x}$ is not always zero; if it were zero, that system would actually not be very interesting. So,

$$m\ddot{x} + kx = 0$$

which is our obtained differential equation. When depending on a single variable, using the conservation of energy in this way is easier than applying Newton's law.

We will work separately with one particular form among all types of energies. That is when the energy is

$$\frac{1}{2}M\dot{q}^2 + U(q) = E$$

Most single-variable dependent energies can actually be written in this way. If there are multiple variables, the energy can sometimes be brought to such a form by replacing the other variable using a conservation law. Differentiating with respect to time as before,

$$M\dot{q}\ddot{q} + \frac{dU}{dq}\dot{q} = 0$$

$$\Rightarrow M\ddot{q} + \frac{dU}{dq} = 0$$

[4] Now we expand $\frac{dU}{dq}$ with respect to the equilibrium point q_0 using the Taylor series-

$$\frac{dU}{dq} = \frac{dU}{dq}\bigg|_{q=q_0} + q\frac{d^2U}{dq^2}\bigg|_{q=q_0} + \cdots$$

Notice again, if q_0 is the equilibrium point, the term $\frac{dU}{dq}\Big|_{q=q_0}$ will be zero, because this term basically represents the negative value of the force experienced by the object at point q_0. Furthermore, if we want to get simple harmonic motion, we will assume the terms with the second and higher powers of q to be zero. Then,

$$M\ddot{q} + q\frac{d^2U}{dq^2}\bigg|_{q=q_0} = 0$$

$$\Rightarrow \ddot{q} + q\frac{U''(q_0)}{M} = 0$$

The frequency of this simple harmonic motion is

$$\omega_0 = \sqrt{\frac{U''(q_0)}{M}} \tag{2}$$

where U'' denotes the second derivative of U with respect to q. To prove the same formula, we could have initially expanded U using the Taylor series. But keep in mind, to get force from potential energy, you have to differentiate once; this reduces the power of the variable by one. So to get the force up to the first power, the energy has to be expanded up to the second power.

The specialty of this formula is that we will not have to take any approximations separately anymore; we have already made the necessary approximations. Let's try to determine the angular frequency of the previous example using this formula. In this case

$$E = \frac{1}{2}m(\dot{r}^2 + \dot{\theta}^2 r^2) - \frac{GMm}{r}$$

We still cannot apply the formula, because another variable $\dot{\theta}$ has remained in the energy. Again, we will replace the variable using the conservation of angular momentum.

$$\begin{aligned} E &= \frac{1}{2}m\left(\dot{r}^2 + \frac{GMR_0}{r^2}\right) - \frac{GMm}{r} \\ &= \frac{1}{2}m\dot{r}^2 + \frac{GMmR_0}{2r^2} - \frac{GMm}{r} \end{aligned}$$

[4] If you look closely at this equation, you can derive a formula yourselves. That is, for a potential energy U, the force experienced by an object is $F = -\frac{dU}{dq}$

Here the first term is kinetic energy. The other terms are only functions of r, so we can call this generalized or effective potential energy.

$$U_{eff} = +\frac{GMmR_0}{2r^2} - \frac{GMm}{r}$$

Then the second derivative of the potential energy is

$$U''_{eff} = \frac{3GMmR_0}{r^4} - \frac{2GMm}{r^3}$$

Since we know the equilibrium point will be $r = R_0$, then $U''_{eff}(R_0) = \frac{GMm}{R_0^3}$. Now using the formula,

$$\omega_0 = \sqrt{\frac{GM}{R_0^3}}$$

which matches our previous answer.

3 Determining the Equation of Motion

In this section, we will solve the equation of simple harmonic motion using our knowledge of solving differential equations. The equations we will work with in this section will be of the form

$$\ddot{x} + \omega_0^2 x + c = 0$$

where c is a constant. We will have no problem with the constant. We can assume a new variable z such that $z \equiv x + \frac{c}{\omega_0^2}$. Then our equation will be

$$\ddot{z} + \omega_0^2 z = 0$$

As we saw at the beginning of the chapter, the general solution of such an equation is complex. This is because, if some real function $f(t)$ satisfies this equation, so will the function $if(t)$. However since the equation doesn't have any cross terms in x or its derivaties, we can initially we will take the entire equation to a complex variable $\tilde{z} = z + iy$, so that $Re(\tilde{z}) = z$. Here, z is the real solution that we are looking for, and y is also a real function. In that case,

$$\ddot{\tilde{z}} + \omega_0^2 \tilde{z} = 0 \rightarrow (\ddot{z} + \omega_0^2 z) + i(\ddot{y} + \omega_0^2 y) = 0$$

For a complex number to be zero, both the real component and the imaginary component have to be zero. This means, if we find the solution for $\tilde{z}$, both $Re(\tilde{z})$ and $Im(\tilde{z})$ are valid real solutions of the equation. In the following examples, you will notice that $Im(\tilde{z})$ and $Im(\tilde{z})$ are linearly dependent, so taking only one of them suffice. We will stick to $Re(\tilde{z})$ to denote the real solution we are looking for. We have seen the solution to the general equation earlier,

$$\tilde{z} = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$$

Here we must remember that z is indicating the position of the object, which is completely real. So the z we are looking for is actually $Re(\tilde{z})$ or just the real part of the variable z. To find the real part of the above equation, we will first take everything into Cartesian coordinates. Since A and B are two possibly complex numbers, let $A \equiv a+ib$ and $B \equiv c+id$, where a,b,c and d are all real numbers. Also, if we expand the exponential part with Euler's formula,

$$\tilde{z} = (a+c)\cos\omega_0 t - (b+d)\sin\omega_0 t + i(b+d)\cos\omega_0 t + i(a+c)\sin\omega_0 t$$

Therefore, $Re(\tilde{z}) = (a+c)\cos\omega_0 t - (b+d)\sin\omega_0 t$. If $a+c = C_1$ and $b+d = C_2$[5] are assumed, then

$$Re(\tilde{z}) = C_1\cos\omega_0 t + C_2\sin\omega_0 t$$

From now on, we will denote the original variable z (including the complex part) by $\tilde{z}$ and its real part, which we want, by z.

The equation we have got for z is the most general form. However, some other different forms are found in various books. You can easily prove these if you know some trigonometry formulas. Some widely used forms are-

$$\begin{aligned} z &= C_1\cos\omega_0 t + C_2\sin\omega_0 t \\ &= C_3\cos(\omega_0 t + \phi_1) \\ &= C_4\sin(\omega_0 t + \phi_2) \end{aligned}$$

By now you might feel, why are we putting so much effort into solving such a simple equation; many of you might have easily solved this equation by integrating it in your higher secondary physics book. But the reason for solving it in so much detail is that what we are learning here, we will also use in the next section to solve damped and forced simple harmonic motion, which cannot be done just by separating variables and integrating.

We have to remember one part from the above derivation; if a term like $z = Ae^{i\omega_0 t}$ comes up, then it will be $Re(z) = C_1\cos\omega_0 t + C_2\sin\omega_0 t$. Definitely remember that A can be a complex number. If you consider A as real, then you will get $Re(z) = A\cos\omega_0 t$, which is wrong.

Example: Solve equation 1. Given that at t=0, the earth's position is at $r = R_0$ and its velocity towards the sun is 5m/s.
Solution: Writing equation 1 in standard form

$$\ddot{x} + \omega_0^2 x = 0$$

where $\omega_0 = \sqrt{\frac{GM}{R_0^3}}$. Its solution will be

$$x = C_1\cos\omega_0 t + C_2\sin\omega_0 t$$

Since our original variable is r and we assumed $r = R_0 + x$,

$$r = R_0 + C_1\cos\omega_0 t + C_2\sin\omega_0 t$$

Substituting $r(t=0) = R_0$ here,

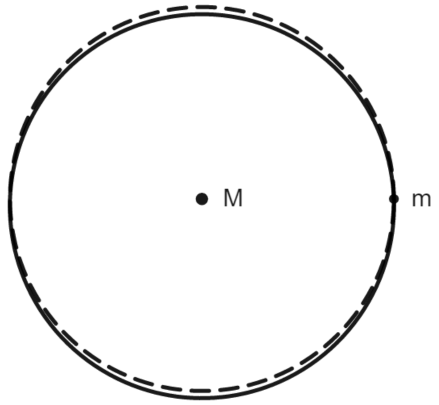
$$R_0 = R_0 + C_1$$

That is, $C_1 = 0$. Again, the velocity being 5m/s towards the sun means $\dot{r}(t=0) = -5m/s$, negative because in polar coordinates the outward direction of r is taken as positive. Then,

$$-5 = \omega_0 C_2\cos(\omega_0\cdot 0)$$

$$\Rightarrow C_2 = -\frac{5}{\omega_0}$$

[5] There is no problem with this. Our final answer must definitely also have two unknown constants, there is no problem if they are different from the initially assumed constants.



Therefore, the final answer stands as

$$r = R_0 - \frac{5}{\omega_0} \sin \omega_0 t$$

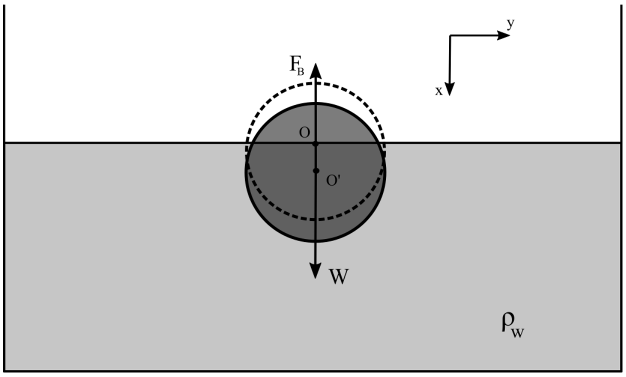
where $\omega_0 = \sqrt{\frac{GM}{R_0^3}}$.

Example: A sphere of radius R is floating in water with half of its volume submerged. If the sphere is given a slight downward push and released, the sphere oscillates in simple harmonic motion. Determine the angular frequency of the motion and the position with respect to time. Given that, the initial displacement of the sphere is x_0 meters downwards from the water surface and the density of water is ρ_w. Ignore surface tension, viscosity of water, and the motion of the water surface.
Solution: To solve this problem, having the knowledge of buoyancy from 9th-10th grade is actually enough. First, let's recall Archimedes' principle; any object submerged in a fluid experiences a buoyant force outwards from the liquid, and its magnitude is equal to the weight of the liquid displaced by the object. Since initially the object is floating in equilibrium, the total force on it is zero. Also, if its volume is V and density is ρ, the buoyant force will be $\frac{V\rho_w g}{2}$ (since half of the total volume was submerged). Then,

$$V\rho g = \frac{V\rho_w g}{2}$$

$$\Rightarrow \rho = \frac{\rho_w}{2}$$

Now let's try to determine what happens after the sphere is pushed downwards and released. Taking the water surface as $x = 0$ and downwards as positive, let's draw the free body diagram of the sphere.



Applying Newton's second law in the vertical direction,

$$\begin{aligned} m\ddot{x} &= mg - \rho_w g V' \\ &= mg - \rho_w g\Big(\frac{V}{2} + \pi R^2 x\Big) \end{aligned}$$

Here since the displacement x is very small, we can imagine the extra submerged volume as a cylinder of height x and cross-sectional area πr^2. So the volume of this water is $\pi R^2 x$. Writing the equation a bit more organized,

$$m\ddot{x} + \pi\rho_w R^2 g x = mg - \frac{V\rho_w g}{2}$$

Just a while ago we saw, the weight of the object is equal to the initial buoyancy value. Therefore, the right side of the equation becomes zero. Now, after dividing the equation by m, expressing m by density and volume,

$$\ddot{x} + \frac{3\pi R^2 \rho_w g}{4\pi R^3 \rho g} x = 0$$

$$\Rightarrow \ddot{x} + \frac{3g\rho_w}{4R\rho} x = 0$$

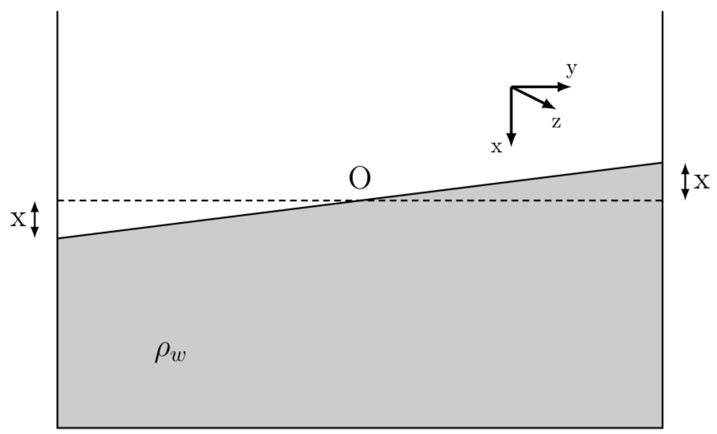
We have previously determined, $\rho/\rho_w = 0.5$. So,

$$\ddot{x} + \frac{3g}{2R} x = 0$$

So, the angular frequency of the motion is $\omega_0 = \sqrt{\frac{3g}{2R}}$ and from here we can directly say that the solution for x will be of the type $x = A\cos\omega_0 t$.[6] Since the initial position is $x = x_0$, so

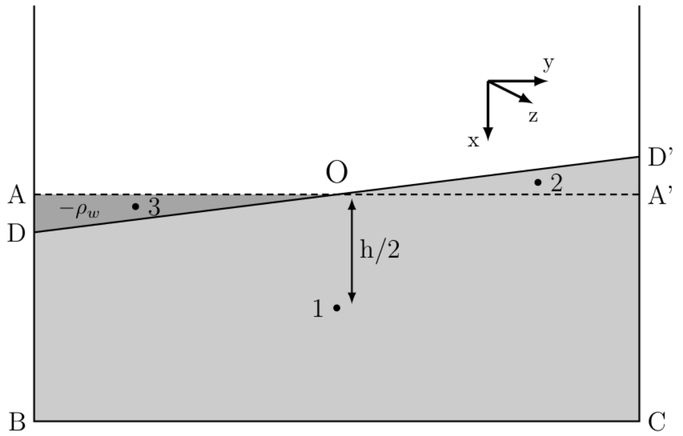
$$x = x_0 \cos\sqrt{\frac{3g}{2R}} t$$

Example:[IPhO 1984] Water kept in a vessel like the figure is oscillating in simple harmonic motion. The water surface is always flat and its midpoint remains fixed. The lengths of the vessel along the x, y, and z (outwards from the page) axes shown in the figure are h, l, and w respectively. If the midpoint of the vessel along the y-axis is always fixed and the value of x is very small, determine the angular frequency of the simple harmonic motion.



[6] If you think a bit, you will see, if initial velocity is zero the coefficient of the sine term is always zero; and if initial position is zero the cosine term is zero. If neither is zero then the coefficients of both are non-zero.

Solution: The hardest part of this problem is figuring out how to start. First look, if we try to apply Newton's second law, we will fall into a very complex situation; we will need to know the amount of force one part of the water is applying on another part. Since we cannot advance much this way, let's rather determine the total energy.



চিত্র 1: In the figure 1, 2, and 3 are the centers of mass of ABCA', OA'D', and OAD respectively. Here the part OABCD'O has water of density ρ_w and the region OAD has water of density $-\rho_w$.

We will take the midpoint O of the vessel as the center of coordinates, the x and y axes will be like the figure. Since this is a large system, its kinetic energy will be the kinetic energy of its center of mass. Earlier we have seen that if the base of a right-angled triangle is a and height is b, the coordinates of its center of mass with respect to the intersection point of the base and hypotenuse is $(2a/3, b/3)$. The volume of triangles OAD and OA'D' is $V = \frac{1}{2} \cdot \frac{l}{2}xw$ (AD = A'D' = x). The center of mass of water up to height h is at the point $(h/2, 0)$. In that case, if the center of mass of the system is (x_{cm}, y_{cm}),

$$
\begin{aligned}
Mx_{cm} &= M\frac{h}{2} + \rho_w V \frac{-x}{3} + (-\rho_w V)\frac{x}{3} \\
\Rightarrow \rho_w lhwx_{cm} &= \frac{\rho_w lhwh}{2} - \frac{\rho_w lwx^2}{6} \\
\Rightarrow x_{cm} &= \frac{h}{2} - \frac{x^2}{6h}
\end{aligned}
$$

Similarly in the y-axis,

$$
\begin{aligned}
My_{cm} &= (M \cdot 0) + \rho_w V \frac{2 \cdot l}{3 \cdot 2} + (-\rho_w V)\frac{2 \cdot -l}{3 \cdot 2} \\
\Rightarrow \rho_w lhwy_{cm} &= \frac{\rho_w l^2 wx}{6} \\
\Rightarrow y_{cm} &= \frac{lx}{6h}
\end{aligned}
$$

If the mass of the entire system is written by M, the total energy of the system will be the kinetic energy of the center of mass and the gravitational potential energy of the center of mass; that is,

$$E = \frac{1}{2}M(\dot{x}_{cm}^2 + \dot{y}_{cm}^2) - Mgx_{cm}$$

Differentiating the obtained x_{cm} and y_{cm} with respect to time,

$$\dot{x}_{cm} = -\frac{x\dot{x}}{3h}$$

$$\dot{y}_{cm} = \frac{l\dot{x}}{6h}$$

Substituting the values, the total energy-

$$E = \frac{1}{2}M\left(\frac{x^2\dot{x}^2}{9h^2} + \frac{l^2\dot{x}^2}{36h^2}\right) + \frac{Mgx^2}{6h} - \frac{Mgh}{2}$$

The energy hasn't quite become like simple harmonic motion yet; but notice, out of the two parts of kinetic energy, one has $x^2\dot{x}^2$, the other has only $\dot{x}^2$. If x is very small, then the first term is negligible compared to the second. Writing without this term,

$$E = \frac{1}{2}\frac{Ml^2}{36h^2}\dot{x}^2 + \frac{1}{2}\frac{Mg}{3h}x^2 - \frac{Mgh}{2}$$

This is an energy equation of a single variable; now if you want you can determine the equation of motion by differentiating it with respect to time like before; or you can notice that the equation is like the energy of a spring whose mass is $\frac{Ml^2}{36h^2}$ and spring constant is $\frac{Mg}{3h}$. In both cases, the angular frequency will come out to be

$$\omega_0 = \sqrt{\frac{12gh}{l^2}}$$

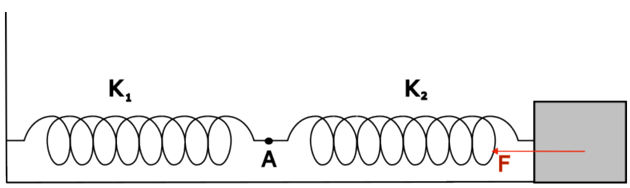
4 Spring Connections

A large part of simple harmonic motion problems is about systems made of several springs. In this section, therefore, we will discuss systems made of springs. Before discussing, we need to know two things-

1. If a spring system applies a force $-ax$ on an object, where x indicates the length extension of the system, then we call a the equivalent spring constant k_{eff} of the system. In this case, we can replace the entire spring system with a single spring of spring constant k_{eff}.

2. In most cases, the mass of the spring is completely negligible; in that case, the net force acting on the spring is zero.

Series Connection: We basically get a series connection if another spring is connected to one end of a spring. In the figure below, two springs with spring constants k_1 and k_2 are connected in series.



চিত্র 2: Two springs with k_1 and k_2 spring connected in series

Since the spring is massless, the net force at any point of it will be zero. Let the first spring be extended by x_1 from equilibrium and the second spring be extended by x_2. In this case, the net force at point A: (assuming rightwards as positive)

$$F_{net,A} = k_2x_2 - k_1x_1 = 0$$

$$\Rightarrow k_2x_2 = k_1x_1$$

That means the extensions of the two springs depend on each other! Again, only the second spring is applying force on the block. That is, $F = -k_2x_2$. If we want to replace these two springs with one spring, then $F = -k_2x_2 = -k_{eff}(x_1 + x_2)$ has to be true.[7] Assuming $x_1 + x_2 = X$ and substituting the value of x_2, it can be shown

$$F = -\frac{k_1k_2}{k_1 + k_2}X$$

$$\Rightarrow k_{eff} = \frac{k_1k_2}{k_1 + k_2}$$

$$\Rightarrow \frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Therefore, if two or more springs are connected in series, then to determine their equivalent spring constant, spring constants have to be added like resistances in parallel!

Example: A spring with spring constant k is cut in a ratio of $a : b$. Determine the spring constants k_a and k_b of each part.

Solution: We will need two equations to determine the values of two variables. Determining the first one is easy; if we connect k_a and k_b in series we will get back the full spring, whose spring constant is k. Therefore,

$$\frac{1}{k_a} + \frac{1}{k_b} = \frac{1}{k}$$

Getting the second equation is comparatively a bit hard. For this, we basically have to assume that the extension of the spring is uniform, which is always true for massless springs. Suppose, we extended the original spring by length l. In this case, the extension of the a proportion part will be $\frac{al}{a+b}$ and the extension of the b proportion part will be $\frac{bl}{a+b}$. In that case, again equating the force at the point where they are joined:

$$k_a\frac{al}{a+b} = k_b\frac{bl}{a+b}$$

$$\Rightarrow k_aa = k_bb$$

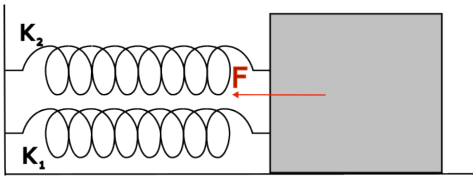
Solving the two obtained equations, it can be shown

$$k_a = \frac{a+b}{a}k \quad \text{and} \quad k_b = \frac{a+b}{b}k$$

That means the spring constant of the smaller part will be greater than the spring constant of the larger part!

Parallel Connection: If the ends of two springs stay at the same position, then that will be a parallel connection.

[7]Since force F works to displace a total distance of $(x_1 + x_2)$ from equilibrium, this displacement has to be kept equal.

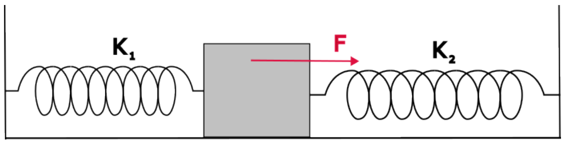


চিত্র 3: Two springs with k_1 and k_2 springs in parallel connection

In this case, both springs apply force to the object. If the displacement of the entire block is x, then $F = -k_1x - k_2x$. That is, in parallel connection

$$k_{eff} = k_1 + k_2$$

Notice in this case if the displacement of the block is x the torque on the block does not become zero. The forces applied by the two springs are k_1x and k_2x respectively, which are not equal. In this case usually the block bends. However, we have assumed that by applying extra torque the extension of both springs has been kept at x. Compared to this, a more ideal parallel connection could be called the connection below:



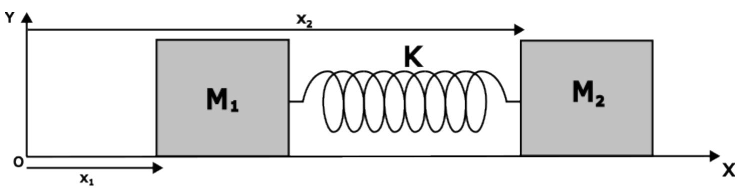
চিত্র 4: Parallel connection created by two springs on both sides of a block

Suppose the block has been moved distance x to the left from equilibrium. In that case, the total force acting on the block

$$\boldsymbol{F} = -k_1\boldsymbol{x} - k_2\boldsymbol{x}$$
$$\Rightarrow \boldsymbol{F} = -(k_1 + k_2)\boldsymbol{x}$$

Therefore, in this case as long as the displacement is parallel to the ground, the equivalent spring constant is $k_{eff} = k_1 + k_2$. For any complex connection, if solved by determining the total force on the object, the behavior of the system is well known. Unless it is a very simple system, it is better not to use the series-parallel connection formulas.

Miscellaneous: In most two-body systems, using the conservation of momentum, it can be mathematically reduced to an identical one-body system. For example, we can work with the system below:



চিত্র 5: Two objects connected by a spring

We want to determine how this system changes with time. Let the position of M_1 be x_1 and the position vector of M_2 be x_2. In that case, the equations of motion:

$$M_1\ddot{x}_1 = -k(x_1 - x_2 + l_0) \tag{3}$$

$$M_2\ddot{x}_2 = -k(x_2 - x_1 - l_0) \tag{4}$$

where l_0 is the length of the spring in its equilibrium state. These two equations contain all the information to solve the system. Adding the two equations-

$$M_1\ddot{x}_1 + M_2\ddot{x}_2 = 0$$

This equation is the equation of conservation of momentum! Multiplying equations 6.3 and 6.4 by M_2 and M_1 respectively and subtracting the obtained equations, we can get another equation-

$$M_1M_2(\ddot{x}_1 - \ddot{x}_2) = -k(M_1 + M_2)(x_1 - x_2 + l_0)$$

Now, let $(x_1 - x_2 + l_0) = z$, which basically indicates the extension of the spring! Then,

$$\frac{M_1M_2}{M_1 + M_2}\ddot{z} = -kz$$

Looking at the equation you might have caught that this is basically the equation of motion of an object connected with a spring of spring constant k and mass $M_{eff} = \frac{M_1M_2}{M_1+M_2}$! That is, we can say the frequency of motion will be $\omega = \sqrt{\frac{k}{M_{eff}}}$. That is,

$$x_1 - x_2 = A\cos(\omega t + \phi) - l_0$$

Again, if we take the center of mass as the reference frame, then $M_1x_1 + M_2x_2 = 0$. Using these two we will be able to determine x_1 and x_2.

5 Damped Oscillation

In the simple harmonic motion we discussed earlier, we ignored air resistance or any variable force.[8] We will work with the effect of a resistive force of the form $\boldsymbol{F} = -b\boldsymbol{v}$ dependent on velocity for simple harmonic motion. This type of resistive force is common for simple harmonic motion.

Let's return to our spring-block system again; but this time a resistive force will act on the block whose value will be $F = -2m\gamma\dot{x}$, where γ is a constant. In this case, applying Newton's second law-

$$m\ddot{x} = -kx - 2m\gamma\dot{x}$$

$$\Rightarrow \ddot{x} + 2\gamma\dot{x} + \omega_0^2x = 0$$

where we have assumed $\frac{k}{m} \equiv \omega_0^2$ again. We have already solved this type of equation before;

$$\tilde{x} = e^{-\gamma t}(Ae^{i\sqrt{\omega_0^2-\gamma^2}t} + Be^{-i\sqrt{\omega_0^2-\gamma^2}t})$$

Just like before again, we are only concerned with the real part of $\tilde{x}$. Then,

$$Re(\tilde{x}) = x = e^{-\gamma t}Re(Ae^{i\sqrt{\omega_0^2-\gamma^2}t} + Be^{-i\sqrt{\omega_0^2-\gamma^2}t}) \tag{5}$$

Here basically 3 cases are possible-

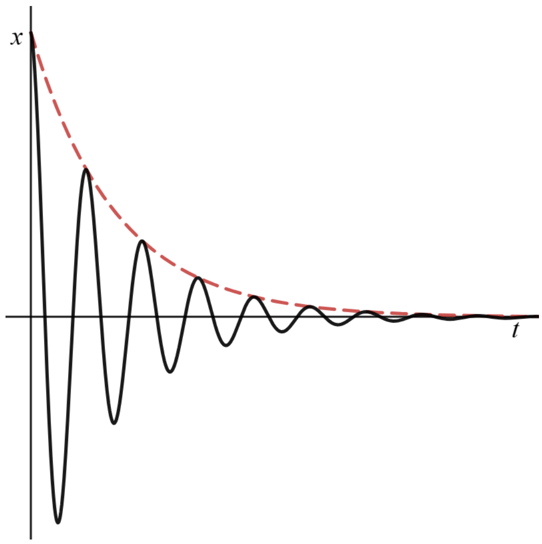
[8]Notice, if a constant force acts, then we can define a new variable and apply the previously discussed part.

5.1 Light damping $\omega_0 > \gamma$

In this case, there is a positive number inside the square root in the exponent. In this case, what will be the real part of the larger section, we have already learned that in section 6.4. Comparing with that,

$$\begin{aligned} x &= e^{-\gamma t} Re(Ae^{i\sqrt{\omega_0^2-\gamma^2}t} + Be^{-i\sqrt{\omega_0^2-\gamma^2}t}) \\ &= e^{-\gamma t}\Big(C_1 \cos(\sqrt{\omega_0^2-\gamma^2}t) + C_2 \sin(\sqrt{\omega_0^2-\gamma^2}t)\Big) \end{aligned}$$

Looking at the equation, it can be said that it is moving in simple harmonic motion, whose amplitude is exponentially decreasing. Besides, its frequency is $\omega = \sqrt{\omega_0^2 - \gamma^2}$, which is less than its natural frequency ω_0; so this motion is called under-damped motion. Such motion is seen in a simple pendulum or spring-mass system due to air resistance.



চিত্র 6: Position vs time graph of a particle when $\omega_0 > \gamma$. The broken line shows the amplitude of motion, which decreases at an exponential rate.

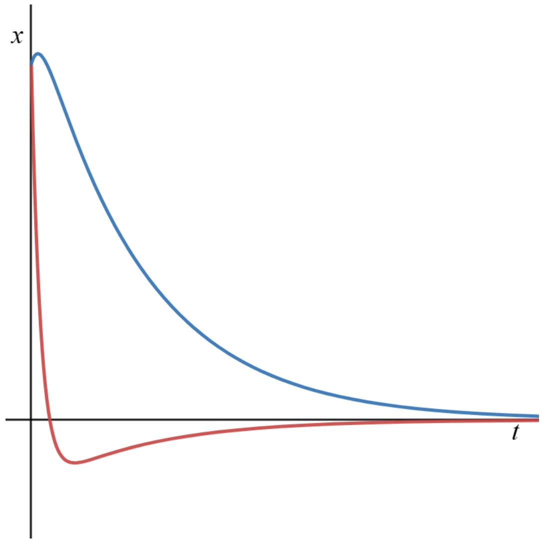
5.2 Heavy damping $\omega_0 < \gamma$

In this case, the part inside the square root also becomes imaginary. In that case, it can be written-

$$Re(\tilde{x}) = x = e^{-\gamma t} Re(Ae^{-\sqrt{\gamma^2-\omega_0^2}t} + Be^{\sqrt{\gamma^2-\omega_0^2}t})$$

$$x = e^{-\gamma t}\left(Ae^{-\sqrt{\gamma^2-\omega_0^2}t} + Be^{\sqrt{\gamma^2-\omega_0^2}t}\right)$$

In this case, no oscillation is seen in the motion, rather the position of the object decreases exponentially and goes to equilibrium.

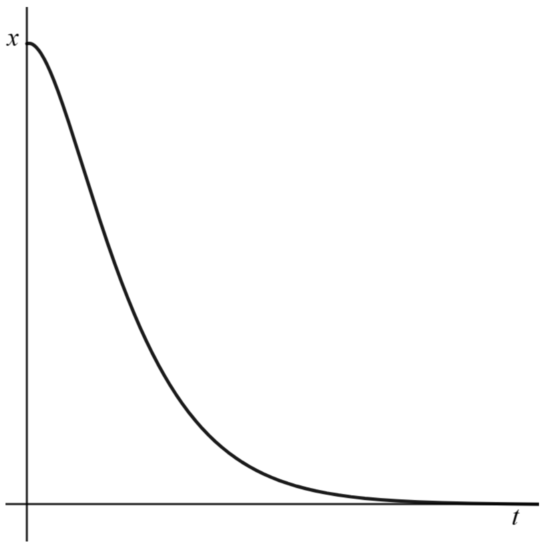


চিত্র 7: Position vs time graph of a particle when $\omega_0 < \gamma$. Two curves are obtained for different initial conditions

To get such damping, the resistance has to be very high. The motion of a simple pendulum or spring-mass system in honey or any such highly viscous medium will be like this.

5.3 Critical damping $\omega_0 = \gamma$

In this case, both terms in the power are equal and zero. Therefore, $x = Ae^{-\gamma t}$ Now we have fallen into a problem. If $\omega_0 = \gamma$ then we have only one solution; which means there is another solution which we have to determine and add with $Ae^{-\gamma t}$. In such a case our second solution will be of the type $tBe^{-\gamma t}$, where B is a constant. You can substitute the solution into the original equation and see that it is indeed a solution. But how the solution came about cannot be shown here without extensive discussion of linear algebra and differential equations. We will not cover it here, but you can differentiate the differential equation with respect to γ and show that if $x(t) = Ae^{-\gamma t}$ is a solution, then $\frac{dx}{d\gamma}$ must also be a solution. The graph below shows the position vs time of a critically damped object. For objects starting in the same initial condition, in critical damping the object returns to equilibrium the fastest without producing any oscillation.



চিত্র 8: Position vs time graph of a particle when $\omega_0 = \gamma$

Example 1 (Light Damping): In a spring-mass system, the mass of the block is m, the spring constant is k and the air resistance force constant is b. The system is pulled a length l from equilibrium and released, resulting in the object moving in oscillatory motion. (a) After how much time will the amplitude of its oscillation become half of the initial amplitude? (b) How many full oscillations will the block complete in this time?

Solution: According to our equation $2m\gamma = b$, that is $\gamma = \frac{b}{2m}$. Again, $\omega_0 = \sqrt{\frac{k}{m}}$. Since it is said that the object is moving in oscillatory motion, it must be an Under-damped (or light damping) system. We know that for light damping, the amplitude decreases exponentially with time: $A(t) = A_0 e^{-\gamma t}$. If $A(t) = \frac{A_0}{2}$,

$$\begin{aligned}\frac{A_0}{2} &= A_0 e^{-\gamma t} \\ e^{\gamma t} &= 2 \\ t &= \frac{\ln 2}{\gamma} = \frac{2m \ln 2}{b}\end{aligned}$$

The new angular frequency of oscillation in the damped state:

$$\omega = \sqrt{\omega_0^2 - \gamma^2} = \omega_0 \sqrt{1 - \frac{b^2}{4mk}}$$

Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{\omega_0\sqrt{1-\frac{b^2}{4mk}}}$. Therefore, the number of full oscillations completed in this time,

$$n = \frac{t}{T} = \frac{2m\ln 2 \cdot \omega_0\sqrt{1-\frac{b^2}{4mk}}}{b \cdot 2\pi} = \frac{\ln 2}{\pi\alpha}\sqrt{1-\frac{\alpha^2}{4}} \qquad \text{where} \qquad \alpha^2 = \frac{b^2}{mk}$$

Example 2 (Critical Damping): Think of a car's suspension system. The mass at each wheel part of the car is M and the spring constant is k. So that the car does not oscillate after hitting a pothole on the road and returns to equilibrium as fast as possible (so that passengers do not feel jolts), what should be the value of the damping constant b of the suspension's shock absorber?

Solution: For the car not to oscillate and to return to equilibrium as fast as possible, the system has to be critically damped. We know, the condition for critical damping is:

$$\gamma = \omega_0$$

Since $\omega_0 = \sqrt{\frac{k}{M}}$ and $\gamma = \frac{b}{2M}$, we can write:

$$\begin{aligned}\frac{b}{2M} &= \sqrt{\frac{k}{M}} \\ b &= 2M\sqrt{\frac{k}{M}} = 2\sqrt{kM}\end{aligned}$$

6 Driven Damped Oscillation

Suppose as before, there is an air resistance force $\vec{F}_{drag} = -2\gamma m\vec{v}$ on a pendulum of natural frequency ω_0. Also, the pendulum is being vibrated by an external agent with a force $mf_0\cos(\omega_d t)$. In that case, the equation of motion:

$$m\ddot{x} = -m\omega_0^2 x - 2\gamma m\dot{x} + mf_0\cos(\omega_d t)$$

$$\Rightarrow \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f_0\cos(\omega_d t)$$

Here we will call ω_d the driving frequency. As before, we will take the entire equation to a complex variable $\tilde{x}$. In this case, as the complex part of $f_0\cos(\omega_d t)$, we will take $f_0 e^{i\omega_d t}$. Since taking its real part will give us back the previous equation, we are not losing any information. Then the equation stands

$$\Rightarrow \ddot{\tilde{x}} + 2\gamma\dot{\tilde{x}} + \omega_0^2\tilde{x} = f_0 e^{i\omega_d t}$$

Since this equation is inhomogeneous, one of its solutions will be extra: the solution that would exist if the equation were homogeneous, and the second one is a function of time due to being inhomogeneous. If the right side were zero, we have already solved the resulting equation earlier. Therefore, its general solution will be

$$\tilde{x} = e^{-\gamma t}(Ae^{i\sqrt{\omega_0^2-\gamma^2}t} + Be^{-i\sqrt{\omega_0^2-\gamma^2}t}) + G(t)$$

We have to determine this $G(t)$ function. There is no general rule to determine such inhomogeneous terms. However, in this case, we will be able to determine this function by applying some real-life knowledge!
Notice, $e^{-\gamma t}$ is multiplied with the first term of $\tilde{x}$. For a fairly large t, the value of $e^{-\gamma t}$ will be very close to zero. Therefore, after a lot of time the object's motion will be $\tilde{x} = G(t)$! That is, in reality, if we keep vibrating a block attached to a spring with force $mf_0\cos(\omega_d t)$, then after a long time its motion will be $G(t)$. Now, if we apply a force of frequency ω_d to an object for a long time, then a very common assumption can be made that the object's motion will also vibrate at frequency ω_d. Then we assume a solution: $G(t) = Ae^{i\omega_d t}$. Substituting this solution into the equation

$$-\omega_d^2 Ae^{i\omega_d t} + 2\gamma\omega_d iAe^{i\omega_d t} + \omega_0^2 Ae^{i\omega_d t} = f_0 e^{i\omega_d t}$$

$$\Rightarrow A = \frac{f_0}{\omega_0^2 - \omega_d^2 + 2i\omega_d\gamma}$$

That is, the equation of the object's position after a long time:

$$x = Re(\tilde{x}) = Re\left(\frac{f_0 e^{i\omega_d t}}{\omega_0^2 - \omega_d^2 + 2i\omega_d\gamma}\right)$$

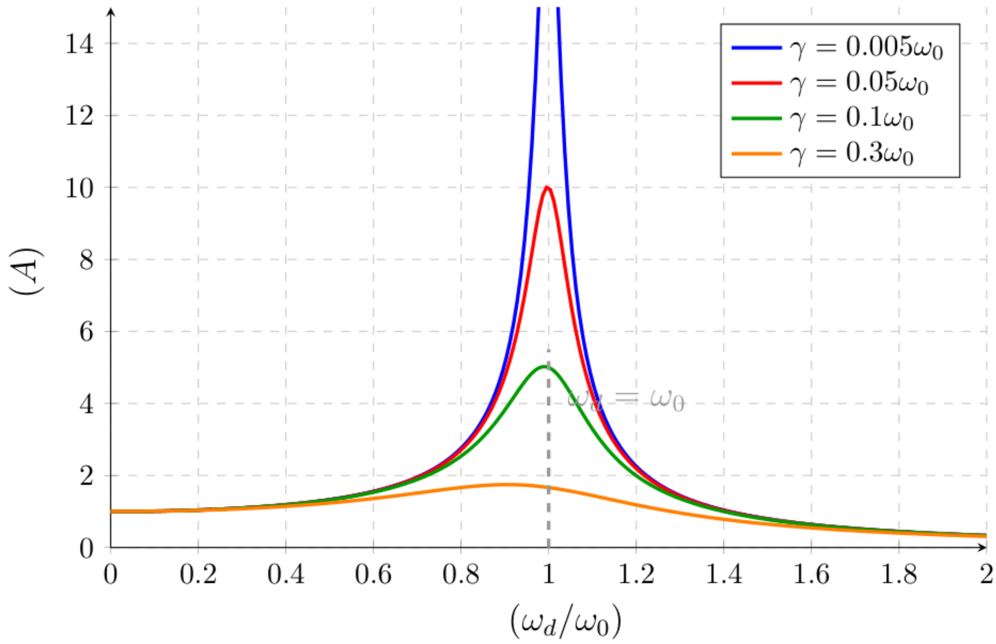
Extracting the real part of this term is somewhat difficult since there is also a complex number in the denominator. We can write the complex number in the denominator in polar form (Appendix B): $\omega_0^2 - \omega_d^2 + 2i\omega_d\gamma = \sqrt{(\omega_0^2-\omega_d^2)^2 + 4\omega_d^2\gamma^2}e^{i\phi}$, where $\phi = \tan^{-1}\frac{2\omega_d\gamma}{\omega_0^2-\omega_d^2}$. Now,

$$Re(\tilde{x}) = Re\left(\frac{f_0 e^{i(\omega_d t-\phi)}}{\sqrt{(\omega_0^2-\omega_d^2)^2 + 4\omega_d^2\gamma^2}}\right) = \frac{f_0\cos(\omega_d t-\phi)}{\sqrt{(\omega_0^2-\omega_d^2)^2 + 4\omega_d^2\gamma^2}}$$

That is, the steady-state amplitude equation of a vibrating object driven by an external periodic force after a long time is:

$$A(\omega_d) = \frac{f_0}{\sqrt{(\omega_0^2-\omega_d^2)^2 + 4\omega_d^2\gamma^2}}$$

Now, if we take a specific spring-mass system (where ω_0 is constant) and a specific value of external force (f_0 is constant), then the amplitude of oscillation A of the object will only depend on two things: the frequency of the driving force (ω_d) and the damping or resistance of the system (γ). First let's try to understand the matter through a graph. Below is a graph showing amplitude (A) versus driving frequency (ω_d) for different values of damping (γ).



A few things are very clear from the graph: there is a maximum value (peak) of amplitude in every curve, and the smaller the value of γ, the closer the maximum point is to $\omega_d = \omega_0$. Let's try to see mathematically what we saw from the graph. Our goal is to determine exactly for which driving frequency (ω_d) the amplitude (A) will be maximum.

For amplitude A to be maximum, the part inside the square root of the denominator of the amplitude equation has to be minimum. Let the part inside the root be D:

$$D(\omega_d) = (\omega_0^2 - \omega_d^2)^2 + 4\omega_d^2\gamma^2$$

According to the rules of calculus, for the minimum value of D, its first derivative has to be zero, that is $\frac{dD}{d\omega_d} = 0$.

$$\begin{aligned} \frac{d}{d\omega_d}\left[(\omega_0^2 - \omega_d^2)^2 + 4\omega_d^2\gamma^2\right] &= 0 \\ 2(\omega_0^2 - \omega_d^2)(-2\omega_d) + 8\omega_d\gamma^2 &= 0 \\ -4\omega_d(\omega_0^2 - \omega_d^2) + 8\omega_d\gamma^2 &= 0 \end{aligned}$$

Since driving frequency $\omega_d \neq 0$, we can write by dividing both sides of the equation by $-4\omega_d$:

$$\begin{aligned} (\omega_0^2 - \omega_d^2) - 2\gamma^2 &= 0 \\ \omega_d^2 &= \omega_0^2 - 2\gamma^2 \\ \omega_d &= \sqrt{\omega_0^2 - 2\gamma^2} \end{aligned}$$

This specific frequency is called the Resonant Frequency and is usually expressed by ω_r. We didn't do the second derivative test here because we have already seen from the graph that the function has only one peak, which is the maximum! That is, resonant frequency:

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$$

At this specific frequency ($\omega_d = \omega_r$), the amplitude of the system becomes maximum. The phenomenon of an object continuing to vibrate at a large amplitude due to the application of an external periodic force on an oscillating system is called Resonance. At resonance, the amplitude of the object is

$$A_{\max} = \frac{f_0}{2\gamma\sqrt{\omega_0^2 - \gamma^2}}$$

You might have noticed a problem here. If there is no resistance in our system ($\gamma = 0$), then the maximum amplitude becomes infinite! Is this actually possible?

Thinking physically, the matter is very logical. When you are pushing exactly at the natural frequency ($\omega_d = \omega_0$) and there is no damping, that means the timing of your push has completely synchronized with the motion of the object. When the object is going to the left, you are pushing to the left, when the object is going to the right, you are pushing to the right. Since there is no resistance, your every push is only adding energy to the system, no energy is being spent. As a result, in every oscillation, the amplitude of the object will keep increasing little by little.

But remember, the amplitude does not suddenly become infinite. For any finite force (f_0), the amplitude increases gradually with time. Mathematically ∞ means that if you apply force for an infinite time, the amplitude will be infinite. But in reality, before reaching a very large amplitude, the spring will cross its elastic limit and break, or the system will collapse. For this reason, resonance can be both necessary and deadly at the same time!

Quality Factor (Q-Factor)

To measure how sharp the peak of amplitude is during resonance, a new quantity called Quality Factor (Q-Factor) is defined. It can be defined by the ratio of the system's ability to store the energy of oscillation to its rate of energy loss. But it can be written very simply through the frequency of oscillation and damping:

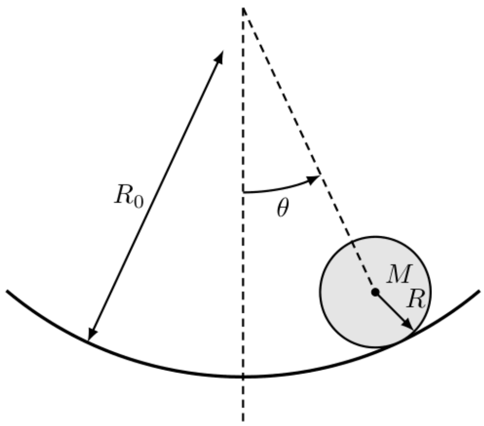
$$Q = \frac{\omega_0}{2\gamma}$$

The higher the value of Q, the lower the damping (γ) in the system will be. As a result, the resonance peak will be very high and very sharp. This means, only if ω_d is very very close to ω_0, then you will get a large amplitude. Even a little deviation will decrease the amplitude a lot. When we tune a channel on a radio or TV, we basically change the natural frequency of the internal circuit to match the frequency of the desired station (cause resonance). Having a high Q-factor is very essential there, so that the circuit only receives the signal of the specific station (high and sharp peak) and cancels out the signals of the adjacent stations.

7 Exercise

7.1 Cylinder rolling on a concave surface

A solid cylinder of mass M and radius R is placed on a fixed concave surface of radius R_0. If the cylinder is slightly displaced from its equilibrium position and released, it rolls without slipping. Determine the time period of small oscillations of the cylinder. (Assume, $R_0 > R$)



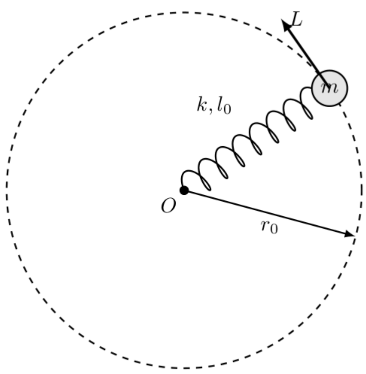
7.2 Dimensional Analysis for Non-linear Oscillators

e know that if $\boldsymbol{F} = -k\boldsymbol{x}$, the motion is simple harmonic and the time period does not depend on the amplitude. But suppose the restoring force acting on an object is proportional to the cube of its displacement, i.e., $F = -cx^3$ (where c is a constant). This is by no means simple harmonic motion. If the mass of the object is m and the amplitude of oscillation is A, show using only Dimensional Analysis, without solving the differential equation of motion, that its time period is inversely proportional to the amplitude (i.e., $T \propto 1/A$).

7.3 Effective Potential and Rotation

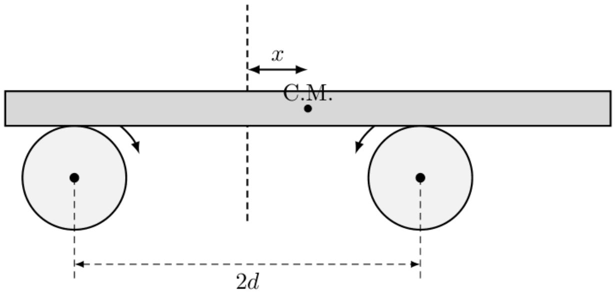
We discussed Effective Potential in Section 2.2. Let's see a great application of it. On a horizontal frictionless table, a particle of mass m is attached to a fixed point on the table by a spring of spring constant k and natural length l_0. The particle is given a push so that it revolves on the table with an angular momentum L. Prove that if the equilibrium radius of the particle is r_0, the frequency of small radial oscillations along this radius will be:

$$\omega = \sqrt{\frac{k}{m} + \frac{3L^2}{m^2 r_0^4}}$$



7.4 Rod on Rotating Cylinders

A uniform horizontal rod of length l and mass m is placed on two rapidly rotating cylinders. The distance between the axes of the two cylinders is $2d$ and they are rotating with equal angular velocity but in opposite directions (such that their upper surfaces move towards the center of the rod). The coefficient of kinetic friction between the rod and the cylinders is μ_k. The rod is slightly displaced horizontally from its equilibrium position and released. Show that the rod will oscillate in simple harmonic motion and determine its time period.



7.5 Rocking Hemisphere

A solid uniform hemisphere of radius R is placed on a flat horizontal frictionless table. (It is good to know that the center of mass of a solid hemisphere is at a distance of $\frac{3}{8}R$ below its flat surface). The hemisphere is tilted by a very small angle from its equilibrium position and released. Determine the time period of its small oscillations.